

The truncation error of a flow step is Gaussian times $e^{2t\|G\|_\infty}$, so the renormalization step is small exactly on small-field configurations

The linearization of the gradient flow is the covariant heat equation with a curvature term,

$$\partial_s \delta B_\mu = D^2 \delta B_\mu + 2[G_{\mu\nu}, \delta B_\nu], \quad D_\mu = \partial_\mu + [B_\mu, \cdot],$$

so the flow Jacobian $D\Phi_t$ is the kernel of that equation. Kato's inequality dominates the covariant heat kernel by the free one pointwise, and the curvature term is a bounded multiplication operator, so Duhamel gives

$$|D\Phi_t(x, y)| \leq e^{2t\|G\|_\infty} \frac{1}{(4\pi t)^{3/2}} e^{-|x-y|^2/(4t)}.$$

Truncating the conjugated Hamiltonian of the flow-conjugation note at range R therefore costs a relative error of order

$$\varepsilon(R, t) \sim \exp \left[2t\|G\|_\infty - \frac{R^2}{4t} \right],$$

and with R of order the scale $\sqrt{8t}$ the exponent is $2t\|G\|_\infty - 2$: the truncation is small **precisely when** $t\|G\|_\infty \lesssim 1$, that is when the flowed field strength is below the inverse square of the scale being integrated out. This is the small-field condition of constructive renormalization group, derived here from the flow rather than assumed, and it explains why Balaban's programme splits configurations into small-field and large-field regions: on small fields the flow truncation is exponentially accurate, and on large fields the Jacobian bound degrades exponentially and no truncation of this kind is available. The flow's own monotonicity controls $\int |G|^2$ and leaves $\|G\|_\infty$ open, so the missing estimate is a pointwise bound on the flowed field strength. Constants explicit; nothing promoted.

Corrected in part. The statements below are continuum statements at fixed physical scale. Within one lattice renormalization step the compactness of the gauge group bounds $a^2\|G\|_\infty$ by π and the flow time of a doubling is $a^2/2$, so the exponential factor is a pure number and the truncation error is uniformly small; see the lattice-truncation note.

1. The linearized flow

Take the continuum flow in the gauge-modified form $\partial_s B_\mu = D_\nu G_{\nu\mu} + \lambda D_\mu \partial_\nu B_\nu$ with $\lambda = 1$ (Lüscher, arXiv:1006.4518v3, equations (1.1)–(1.2) and (2.2); passage level via the local companion).

Proposition 1. The variation δB of a solution obeys

$$\partial_s \delta B_\mu = D^2 \delta B_\mu + 2[G_{\mu\nu}, \delta B_\nu].$$

Proof. Varying the first term,

$$\delta(D_\nu G_{\nu\mu}) = D_\nu(D_\nu \delta B_\mu - D_\mu \delta B_\nu) + [\delta B_\nu, G_{\nu\mu}] = D^2 \delta B_\mu - D_\nu D_\mu \delta B_\nu + [\delta B_\nu, G_{\nu\mu}],$$

and $D_\nu D_\mu \delta B_\nu = D_\mu D_\nu \delta B_\nu + [G_{\nu\mu}, \delta B_\nu]$. Varying the gauge term gives $D_\mu D_\nu \delta B_\nu$ to the same order, which cancels the second piece, leaving $D^2 \delta B_\mu - 2[G_{\nu\mu}, \delta B_\nu] = D^2 \delta B_\mu + 2[G_{\mu\nu}, \delta B_\nu]$. $\square$

So $D\Phi_t$ is the solution kernel of a covariant heat equation with a zeroth-order term given by the curvature acting in the adjoint representation.

2. The Gaussian bound

Proposition 2. Let $\|G\|_\infty = \sup_{0 \leq s \leq t, x} \|G(s, x)\|$ in the adjoint operator norm. Then the kernel of the linearized flow obeys

$$|D\Phi_t(x, y)| \leq e^{2t\|G\|_\infty} K_t^{\text{free}}(x - y), \quad K_t^{\text{free}}(z) = \frac{e^{-|z|^2/(4t)}}{(4\pi t)^{3/2}}.$$

Proof. Write the equation as $\partial_s u = D^2 u + Mu$ with $(Mu)_\mu = 2[G_{\mu\nu}, u_\nu]$, a multiplication operator of norm at most $2\|G\|_\infty$. Duhamel gives $u(t) = e^{tD^2}u(0) + \int_0^t e^{(t-s)D^2}Mu(s)ds$, and iterating, $\|u(t)\| \leq \sum_n \frac{(2t\|G\|_\infty)^n}{n!} \sup \|e^{D^2}\|$ -type bounds; pointwise, Kato's inequality gives $|e^{sD^2}f| \leq e^{s\Delta}|f|$ for the covariant Laplacian in any representation (Simon, J. Funct. Anal. 32 (1979) 97, metadata level), so each term is dominated by the free heat kernel and the series sums to $e^{2t\|G\|_\infty}$. $\square$

The bound is saturated in the free case, where $G = 0$ and $D\Phi_t = e^{t\Delta}$ exactly, in agreement with the free-field note §2.

3. The truncation error

The conjugated kinetic operator of the flow-conjugation note §2 is built from $D\Phi_t$, which couples links at separation $|x - y|$ with weight bounded by Proposition 2. Truncating it to range R discards the tail $|x - y| > R$, whose weight relative to the whole is

$$\varepsilon(R, t) \leq e^{2t\|G\|_\infty} \int_{|z|>R} K_t^{\text{free}}(z) d^3z \leq C e^{2t\|G\|_\infty} e^{-R^2/(4t)} \left(1 + \frac{R}{\sqrt{t}}\right),$$

by the standard Gaussian tail estimate. With the natural choice $R = \sqrt{8t}$, the exponent is $2t\|G\|_\infty - 2$, so

Corollary 3. The flow-truncation step at scale $\sqrt{8t}$ has relative error

$$\varepsilon \lesssim \exp[2t\|G\|_\infty - 2],$$

small when $t\|G\|_\infty \lesssim 1$ and exponentially large otherwise. Taking $R = \kappa\sqrt{8t}$ improves the exponent to $2t\|G\|_\infty - 2\kappa^2$ at the cost of a longer-range effective Hamiltonian, so a fixed accuracy is available whenever $\kappa^2 \gtrsim t\|G\|_\infty$.

4. What the condition says

Write $\ell = \sqrt{8t}$ for the scale being integrated out. The condition $t\|G\|_\infty \lesssim 1$ reads

$$\|G\|_\infty \lesssim \frac{8}{\ell^2},$$

that is, the field strength must be below the inverse square of the scale. In the units of the obligations map the lattice field strength at spacing a is at most of order $1/a^2$, so at the first step, $\ell \sim a$, the condition is marginal; after the field has been smoothed over $\ell \gg a$ the typical $\|G\|$ falls and the condition is comfortable on smooth configurations and violated on rough ones.

This is the small-field condition of constructive renormalization group, obtained here as the condition for the flow truncation to be accurate, and it identifies the difficulty by name:

- on **small-field** configurations, $\|G\|_\infty \leq \eta/\ell^2$ with η small, the truncation error is $e^{-2\kappa^2+2\eta}$ and every step of the route of the flow-conjugation note §6 is controlled;
- on **large-field** configurations the Jacobian bound degrades like $e^{2t\|G\|_\infty}$ and no truncation of this kind is available, so those regions must be handled by a different argument.

The split is exactly the one Balaban's programme makes (Commun. Math. Phys. 122 (1989) 355 and the accompanying series, metadata level), and the derivation above says why it is forced rather than chosen.

5. What the flow itself controls, and what it leaves

The flow decreases the action monotonically, $\frac{d}{ds}S(B_s) = -\|D^*G\|_2^2 \leq 0$ (comparison note §4), so

$$\int |G(t, x)|^2 d^3x \leq \int |G(0, x)|^2 d^3x$$

for all t : an L^2 bound, uniform in the flow time. Proposition 2 needs an L^∞ bound, and the gap between them is the whole question. Two remarks fix its size.

The free case is a parabolic smoothing estimate. For $G = 0$ the linearized flow is the heat semigroup and $\|e^{t\Delta}f\|_\infty \leq (4\pi t)^{-3/4}\|f\|_2$, so an L^2 bound on the initial curvature gives $\|G(t)\|_\infty \lesssim t^{-3/4}\|G(0)\|_2$, which beats the required $1/t$ for large t and fails for small t . The interacting statement of this type is the a priori estimate that a construction must supply.

Monotonicity is not enough by itself. An L^2 bound permits a configuration with curvature concentrated on a small region at arbitrary height, which is precisely a large-field region; excluding it is a statement about the measure in the Euclidean formulation and about the state in the Hamiltonian one. In the Hamiltonian framework there is no integration over configurations to absorb such regions into a small probability, which makes the Hamiltonian route harder at exactly this point than the Euclidean one.

6. Consequence for STATE

The truncation error of a flow-based renormalization step is now explicit: Gaussian in the ratio of the truncation range to the flow radius, multiplied by $e^{2t\|G\|_\infty}$. The step is controlled exactly on configurations with $\|G\|_\infty \lesssim \ell^{-2}$, which derives the small-field condition instead of assuming it. The missing estimate is pointwise control of the flowed field strength, an L^∞ bound where the flow supplies only L^2 , and in the Hamiltonian framework there is no probabilistic route around it. That is the sharpest statement of the remaining obstacle this programme has reached, and it points at the Euclidean formulation as the place where the large-field regions can be given small weight.